

Virtual Tutors for Science Labs: Exploring the Impact of GPT Integration and 3D Tutor Avatars on Student Engagement and Learning Outcomes

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Abstract

In virtual learning environments, pedagogical agents are often used to simulate the presence of an instructor. Traditionally, these agents deliver instructional content through narration, either alone or accompanied by a visual avatar. Recent advancements in large language models, such as GPT, have enabled the development of conversational agents that provide more responsive and interactive learner support. This study examined the impact of integrating GPT and a 3D avatar into conventional voice-guided instructional modules. A 2 (3D avatar: present vs. absent) $\times$ 2 (GPT chat: enabled vs. disabled) factorial experiment was conducted to evaluate their effects on the perceived social presence of the agent, engagement, and learning outcomes—measured by self-reported knowledge gain and quiz performance. Results from a preliminary study with 34 participants showed that GPT integration significantly enhanced social presence, perceived knowledge gain, and quiz scores. Although the agent's avatar did not improve quiz performance, it led to higher perceived knowledge gain and social presence. These findings highlight both the individual and combined effects of GPT and avatars in virtual instructional environments.

CCS Concepts

• **Human-centered computing** $\rightarrow$ **Interaction Design**; *Empirical studies in interaction design*.

Keywords

Pedagogical Agent, STEM Education, Conversational AI, Agent Avatar

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1 Introduction

Over the past decade, remote education has grown substantially, leading many engineering programs to adopt virtual science laboratories [13, 20, 22]. Unlike traditional distance learning—where students passively view recorded lectures—virtual labs offer interactive, simulated hands-on experiences [22]. This approach is particularly valuable in fields like microelectronics fabrication, where access to physical laboratories is limited by the high cost of clean-room facilities—often reaching multi-million dollars—and the safety risks associated with operating complex machinery [23]. Virtual labs help mitigate these challenges by allowing students to engage with realistic digital simulations of advanced equipment, reducing both financial and safety barriers [16].

Despite these benefits, maintaining student engagement in remote lab environments remains a key challenge [24]. In traditional lab settings, teaching assistants (TAs) offer real-time, personalized support that helps students manage tasks and stay engaged [29]. Remote learners, however, often rely on static instructions or delayed communication, which limits immediate feedback and can diminish engagement [24]. To address this, researchers have introduced virtual pedagogical agents designed to simulate real-time instructional support and guide students through lab activities [5, 6]. These agents have been shown to improve learners' sense of social presence and motivation, fostering more engaging and effective learning experiences [9, 21]. More recently, generative AI—particularly large language models such as GPT—has opened new opportunities for delivering dynamic, conversational support in virtual learning environments. Students can now ask context-specific questions (e.g., “What is the name of the chip I’m holding?”) during interactive tasks (e.g., placing a chip on a testbed), creating more personalized and engaging learning experiences.

Although GPT-powered pedagogical agents have received growing attention, most prior research has examined conversational AI

and pedagogical avatars in isolation, with relatively little known about their combined impact [10, 27, 30]. To address this gap, we developed a virtual microelectronics fabrication lab designed to teach the operation of complex equipment—specifically, X-ray tomography—accompanied with a pedagogical agent. This study explores the effects of two key design features: (1) GPT-based conversational support and (2) a 3D pedagogical agent avatar. Through comparative analysis, we examine their individual and combined impact on student engagement and learning outcomes. The findings aim to provide insights for the design of future AI-powered pedagogical agents in virtual science laboratories.

2 Related Works

2.1 AI Conversational Agents and Their Impact on Online Learning

AI conversational agents—especially those powered by large language models such as GPT—have recently gained attention as effective tools for enhancing learning experiences in online environments [14]. Existing research highlights the critical role of these agents in promoting student engagement. For instance, Hamid et al. demonstrated that integrating ChatGPT into process-driven problem-based learning (PBL) led to increased student engagement and a higher frequency of student-generated questions compared to a control group [14]. From a theoretical standpoint, these findings resonate with scaffolding principles in inquiry-based learning, which suggest deeper student engagement when learners are encouraged to ask questions and receive immediate, contextually relevant feedback during problem-solving activities [11, 12]. Unlike static educational materials such as pre-recorded lectures or FAQ sections, AI conversational systems facilitate dynamic, interactive dialogues that allow students to pose real-time questions, reflect on responses, and engage iteratively. Such interactions align closely with active learning theories that emphasize learners' active participation in knowledge construction [7, 18]. Supporting this idea further, Chien et al. found that generative AI used as real-time scaffolding in an online educational game significantly increased learners' experience of flow—a state of deep focus and immersion—due to the immediate feedback and on-demand assistance provided [8]. Collectively, these findings underscore the potential of GPT-powered conversational agents to enhance student engagement and foster deeper learning processes. Drawing from this literature, we hypothesize:

H1. *GPT-enabled conversational support will enhance engagement and learning in the virtual science lab by enabling students to ask in-situ questions.*

2.2 Pedagogical Agent Avatars, Social Presence, and Learning

Pedagogical agents, lifelike virtual characters created to support learners in multimedia learning contexts, have been extensively explored due to their potential for improving student engagement and educational outcomes in online settings [28]. These agents range from simple voice-only narrators to fully interactive, animated avatars capable of simulating human-like behaviors and interactions [9, 15]. The concept of social presence, defined as

the subjective experience of interacting with another social entity within a mediated environment, is central to the effectiveness of pedagogical agents [4]. Prior research has suggested that social presence facilitates positive learning experiences by creating a sense of connectedness and collaborative interaction between learners and pedagogical agents, mirroring genuine human interactions [4, 17].

Early research by Mayer suggests that highly embodied pedagogical agents, featuring realistic gestures, facial expressions, and natural speech patterns, more effectively establish a sense of social partnership and thereby promote deeper cognitive engagement [18]. Similarly, social agency theory indicates that when learners perceive virtual agents as authentic social entities, their motivation and engagement in processing instructional content notably increase [25]. Baylor and Kim further illustrate this principle, demonstrating how avatar realism, expressiveness, and voice characteristics significantly influence learners' perceptions of an agent's authority and approachability, ultimately affecting learning behaviors and outcomes [3]. The recent advancements in conversational AI, particularly large language models like GPT, have further expanded possibilities for dynamic and real-time interactions in virtual learning environments [7, 8, 12, 14]. Overall, existing literature indicates that integrating conversational AI technologies with richly embodied pedagogical avatars could amplify the positive impacts of social presence on learner engagement and educational performance. Based on these foundations, we hypothesize:

H2. *The presence of a 3D avatar will strengthen learners' sense of social presence, further supporting engagement and learning performance.*

3 Method

3.1 Study Context: Virtual Microelectronic Fabrication Lab

This study context focuses on a virtual microelectronic fabrication and physical assurance lab, designed to simulate hands-on training in analyzing microchip quality using X-ray tomography. The virtual lab replicates the typical tasks students perform during in-person lab sessions. The virtual experience first introduces learners to the principles of X-ray tomography, then prompts them to complete a series of interactive procedures—such as opening the X-ray machine door, picking up a sample chip, placing it on the testbed, adjusting machine parameters (e.g., voltage and current), and interpreting the resulting X-ray images of the chip. The virtual pedagogical agent provides step-by-step guidance throughout the process.

3.2 Integration of 3D Pedagogical Agent Avatar and GPT

We employed a 2 (GPT chat: present vs. absent) × 2 (3D agent avatar: present vs. absent) factorial design. The GPT interface was presented in a separate panel adjacent to the instructional panel (see Figure 1), ensuring that all participants received identical scripted instructions across conditions. Participants in the GPT-enabled condition could use a chatbox to ask supplementary questions (e.g., “What is X-ray tomography?”), while core task instructions (e.g., “Now, open the X-ray machine door”) remained consistent

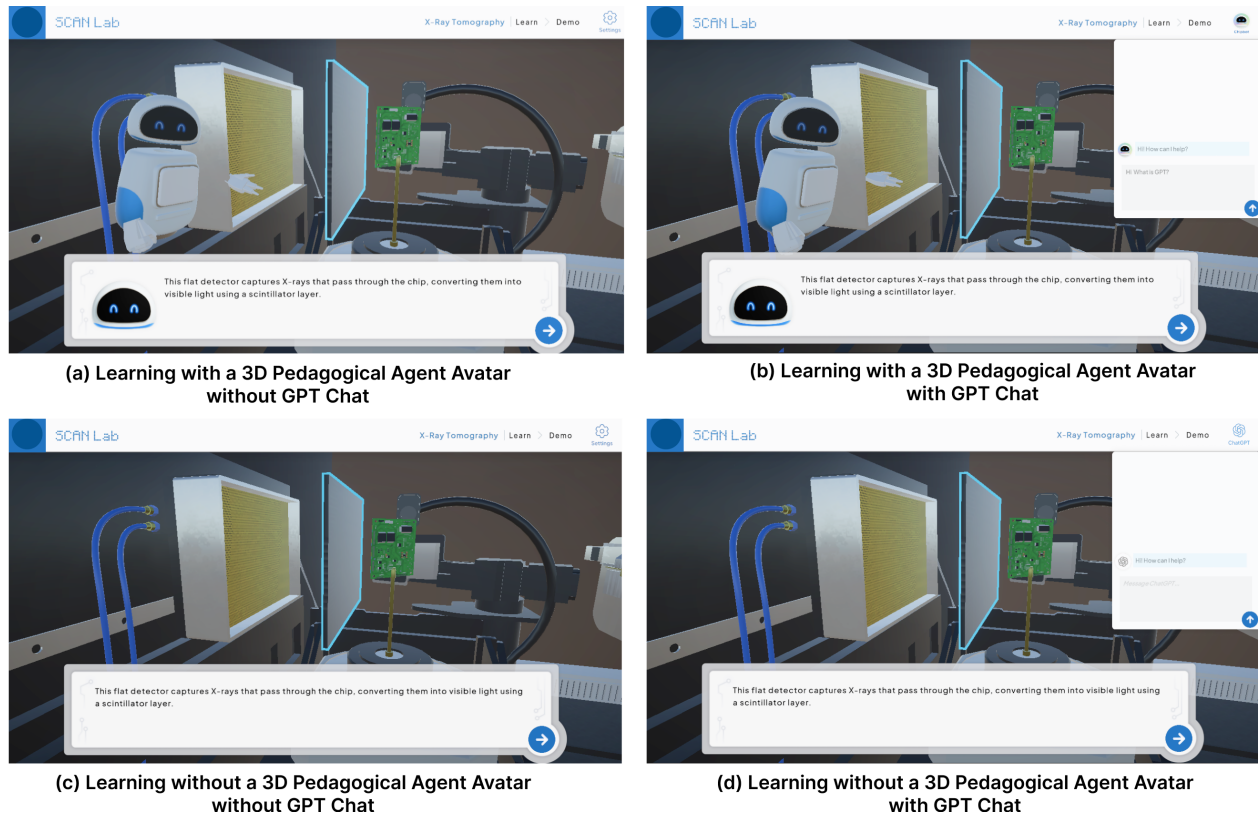


Figure 1: Study overview: four different conditions to investigate the impact of AI chat (powered by GPT) and a 3D avatar.

across all groups. To ensure content relevance, we used a fine-tuned CustomGPT model trained on domain-specific data related to microelectronics and physical assurance. For the 3D agent, a generic robot avatar was selected to avoid engagement biases linked to human-like features (e.g., gender, attire) as documented in prior studies [1]. The avatar's voice was generated using ElevenLabs to provide a consistent auditory experience. The virtual lab was developed in Unity (version 2022.3.40f), and the GPT-3 model was integrated via the OpenAI API.

3.3 Study Design and Participants

A mixed-subjects design was implemented, with GPT presence as a between-subjects factor and 3D avatar presence as a within-subjects factor. To control for time-on-task effects, participants were required to spend a minimum of 5 minutes across all four conditions before proceeding to the survey. On average, participants spent 457.2 seconds in the virtual lab, with no significant time differences observed across conditions. A total of 34 participants were recruited from an online subject pool. Recruitment criteria included individuals with the STEM background and no prior experience in microelectronic fabrication, but an expressed interest in learning about the topic—representing the intended target user group. Participants were randomly assigned to the GPT ($n = 16$) or non-GPT ($n = 18$) condition. Within each group, participants experienced two versions of the lab: one with a 3D agent and one

without. The order of avatar presentation was counterbalanced to minimize order effects. Most participants were aged 25–34 (36%), followed by 18–24 (33%) and 35–44 (26%). Academic backgrounds included engineering (43%), natural sciences such as mathematics or chemistry (13%), and health sciences (16%). None had prior experience operating X-ray tomography equipment, consistent with the recruitment criteria.

3.4 Measures

Social Presence: Social presence was measured using items adapted from Lee et al., including statements such as “I felt as if I was interacting with an intelligent being,” “I felt as if I was learning content from an intelligent being,” “I felt as if the agent was actively responding to me,” and “I felt as if the pedagogical agent and I were engaging in meaningful communication.” Responses were rated on a 7-point Likert scale. The scale demonstrated high internal consistency ($\alpha = 0.949$) [26].

Engagement: Engagement was measured using three items adapted from O'Brien et al. [19]: “This learning experience was fun,” “I was deeply involved in this learning experience,” and “This learning experience was engaging.” The scale showed good internal consistency ($\alpha = 0.798$).

Perceived Knowledge Gain: Participants rated their knowledge before and after the module on three items: “I am knowledgeable about X-ray tomography principles,” “I understand the procedures

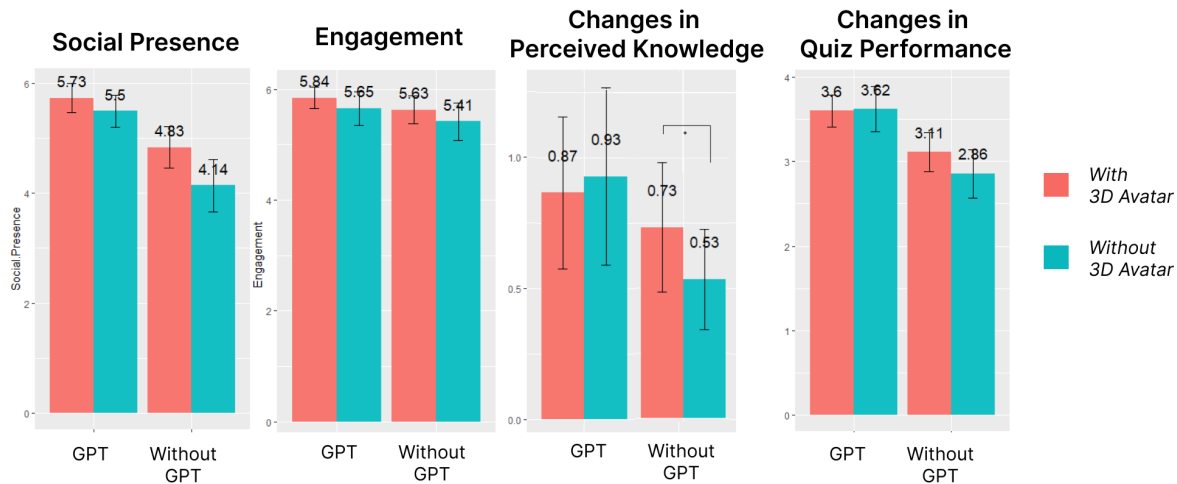


Figure 2: Interaction Between GPT and 3D Avatar on Social Presence, Engagement and Learning Outcomes

involved in operating an X-ray tomography machine,” and “I understand how X-ray source parameters affect image quality.” Ratings were on a 7-point scale. The change in self-assessed scores was used as a dependent variable.

Quiz Performance: Knowledge acquisition was also assessed through pre- and post-module quizzes on the instructional content. The difference in quiz scores was used to evaluate actual knowledge gain.

Other Measures: Usability was assessed using a modified System Usability Scale (SUS) [2], including items such as “I thought this app was easy to use.” Given the potential influence of interface usability on learning outcomes, it was included as a control variable ($\alpha = 0.714$). Demographic information was also collected at the end of the survey.

4 Results

Figure 2 summarizes the results for key dependent variables, including social presence, engagement, perceived knowledge gain, and quiz score change. To evaluate the effects of the experimental manipulations, a series of two-way ANOVAs was conducted, examining the influence of the GPT condition (GPT vs. No-GPT) and the 3D condition (3D avatar vs. No-3D avatar) across these measures. Initial analyses focused on usability perceptions across conditions. Post hoc comparisons revealed no statistically significant differences in usability scores among the four experimental groups. This suggests that participants’ evaluations of the system’s usability were stable and unaffected by the presence or absence of GPT or the 3D avatar.

Social Presence: The analysis of social presence yielded a significant main effect for the GPT condition, $F(1, 55) = 9.60, p < .001$. Participants who interacted with the GPT-enhanced system reported a stronger sense of social presence compared to those in the No-GPT condition. Additionally, the 3D avatar condition demonstrated a marginally significant main effect, $F(1, 55) = 6.47, p = .013$, indicating that the visual presence of an avatar may also contribute to heightened perceptions of social presence.

Engagement: In contrast to social presence, engagement scores did not show significant variation across conditions. Neither the GPT condition ($F(1, 55) = 0.66, p = .419$) nor the 3D condition ($F(1, 55) = 0.53, p = .469$) produced a significant main effect. Furthermore, the interaction between GPT and 3D conditions was non-significant. These results suggest that while GPT and 3D avatars may enhance social presence, they do not necessarily translate into increased engagement within the context of this study.

Changes in Perceived Knowledge: A significant main effect of the GPT condition was observed for perceived knowledge gain, $F(1, 55) = 7.06, p < .01$. Participants in the GPT group reported greater self-assessed learning compared to those in the No-GPT group, indicating that GPT may play a role in enhancing learners’ confidence in their understanding of the material. Although the 3D avatar condition did not yield a statistically significant main effect, post hoc analyses revealed that participants exposed to the 3D avatar reported higher perceived knowledge gain than those without it ($p < .05$) when the GPT interaction was not present.

Quiz Score: Finally, quiz performance was significantly influenced by the GPT condition, $F(1, 55) = 5.01, p < .05$. Participants who interacted with GPT showed a greater increase in quiz scores compared to those in the No-GPT group, suggesting that GPT-enhanced interactions may support learning outcomes. However, the 3D condition did not produce a significant effect on quiz scores, nor was there a significant interaction between GPT and 3D conditions.

5 Discussion

This preliminary study examined the impact of two key features of pedagogical agents in a virtual science laboratory learning environment: (1) the integration of a custom GPT agent to answer students’ in-situ questions during tutorials, and (2) the presence of a 3D avatar representing the pedagogical agent. The findings partially supported our hypotheses, which proposed that both GPT integration and the presence of a 3D avatar would influence perceived social presence, engagement, and learning outcomes. Overall, results indicated that

both the GPT and 3D avatar features enhanced participants' perception of social presence, aligning with prior research on virtual agents in learning environments [3, 25]. Additionally, participants in the GPT condition showed significantly greater improvement in quiz performance compared to those in the non-GPT condition, suggesting that GPT integration positively impacted actual learning outcomes.

In contrast, the presence of the 3D avatar did not yield a significant effect on quiz performance. This may suggest that while the visual presence of an avatar can enhance perceived social presence, it may not directly translate to improved learning performance. However, participants exposed to the 3D avatar reported higher self-assessed knowledge gain than those in conditions without the avatar—particularly when GPT was not integrated. This suggests that while a 3D avatar alone may not improve measurable performance, it can contribute to learners' subjective sense of knowledge acquisition, especially in the absence of interactive AI support. No significant differences in engagement were found across conditions. This may be due to the possibility that engagement was more strongly influenced by the core virtual lab activities rather than the characteristics of the pedagogical agent, as reflected by the similar engagement ratings across all four conditions.

Overall, the findings offer insights into the individual and combined effects of GPT-powered conversational support and 3D avatars in virtual learning environments. They highlight the potential of GPT integration to improve both perceived and actual learning outcomes.

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